# MAHMOUD ABOUZEID

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## SUMMARY

AI & Data Science engineer with hands-on experience building and deploying end-to-end systems across Healthcare AI, Computer Vision, NLP, Retrieval-Augmented Generation (RAG), Mixed Reality, and mobile applications. Developed an explainable clinical risk-prediction pipeline with SHAP interpretability and temporal external validation on NHANES data. Led a 5-member team to deliver an agentic RAG-based AR education platform showcased at AASTMT, which won project funding of 52,250 EGP. Proficient in Python, PyTorch, TensorFlow, scikit-learn, LangChain, FastAPI, Flutter, and Firebase across research, web, and mobile environments.

## TECHNICAL SKILLS

**Languages:** Python, C, JavaScript (Node.js), Dart, HTML, CSS

**Machine Learning & Data Science:** scikit-learn, NumPy, Pandas, SHAP, Feature Engineering, Model Evaluation, Tabular Data Modeling, External Validation, Clinical Risk Prediction, Healthcare AI, Explainable AI

**Deep Learning & Computer Vision:** TensorFlow, PyTorch, Keras, CNNs, U-Net, YOLOv5, OpenCV, TensorFlow Lite, Hybrid CNN + Attention

**NLP & Generative AI:** Transformers, LangChain, Retrieval-Augmented Generation (RAG), Text Summarization, Speech-to-Text, Text-to-Speech, Gemini API

**Backend, Databases & Deployment:** FastAPI, Flask, MongoDB, SQL, FAISS (Vector DB), Firebase Authentication, Cloud Firestore, REST APIs

**Mobile & AR Development:** Flutter, Provider, SharedPreferences, Material 3, fl_chart, Unity (AR/VR), XREAL SDK, TensorFlow Lite (On-Device Inference), Android APK Build

**Data Visualization:** Power BI, Matplotlib, Seaborn, fl_chart, percent_indicator, Interactive Dashboards

**Tools & Workflow:** Git, GitHub, Agile, Jupyter Notebook, Firebase Console, Android Studio, VS Code

## EXPERIENCE

**Freelance AI/ML & Software Developer** | 2025 - 2026 (Part-Time, Intermittent)

- Delivered freelance projects spanning AI/ML, web/mobile development, and data analysis for clients sourced through freelancing platforms and direct referrals.
- Built and deployed machine learning models and data-driven solutions tailored to client requirements using Python, scikit-learn, TensorFlow, and PyTorch.
- Developed web and mobile application features using Flutter and JavaScript/Node.js according to client specifications.
- Performed data cleaning, analysis, and visualization to support client decision-making using Pandas, Power BI, and Matplotlib.
- Managed client communication, project scoping, and delivery independently alongside full-time studies.

## EDUCATION

**B.Sc. in Artificial Intelligence and Data Science**

Arab Academy for Science, Technology, and Maritime Transport (AAST), Egypt | Sept 2021 - Sept 2025 |

**High School Diploma**

Thebes American College, 6th of October City, Giza, Egypt | Sept 2019 - Jul 2021

## PROJECTS

**DualRisk-AI - Diabetes Prediction and Cardiovascular Risk Estimation**

*Python, Pandas, NumPy, scikit-learn, PyTorch, SHAP, Matplotlib, NHANES*

- Developed an explainable healthcare ML pipeline for diabetes prediction and cardiovascular risk-factor burden estimation using structured clinical data from 21,798 NHANES adults across 10 input features.
- Built and compared Logistic Regression, Random Forest, and a compact PyTorch Hybrid CNN + Attention model (8,514 parameters) across two clinical prediction tasks.
- Engineered a reproducible preprocessing pipeline including NHANES file merging, missing value handling, feature engineering, label construction, and stratified train/test splitting.

- Applied SHAP TreeExplainer to Random Forest models, identifying clinically meaningful predictors including HbA1c, glucose, BMI, age, smoking indicators, blood pressure, and cholesterol.
- Performed temporal external validation on NHANES August 2021-August 2023 data (8,153 adults) to evaluate model generalization without retraining.
- Achieved external ROC-AUC of 0.9173 for diabetes prediction (Random Forest) and 0.9521 for cardiovascular risk-factor estimation (CNN + Attention).

**ARAGbMR - Agentic RAG-based Mixed-Reality Education Platform (Graduation Project)**

*Unity, FastAPI, LangChain, RAG, XREAL Air 2 Ultra SDK, Gemini 1.5 Pro, STT/TTS*

- Led a 5-member cross-functional team through full-cycle development of an AR education platform, coordinating backend, AI, and frontend modules using Agile workflows.
- Secured project funding of 52,250 EGP for the graduation project based on the platform's technical merit and presentation.
- Engineered an avatar-based teaching assistant delivering real-time Q&A with an average response latency of 15 seconds, powered by a RAG pipeline integrated with Gemini 1.5 Pro, Speech-to-Text, and Text-to-Speech.
- Developed an automated content pipeline that transforms textbook PDFs (~2,500 pages) into interactive AR slides within 1 hour while maintaining memory usage under 1 GB.
- Deployed and presented the system at AASTMT University, achieving a +26 percentage-point learning gain and user engagement scores of 4.6 out of 5 in evaluation sessions.

**FitLife Pro - Fitness and Nutrition Tracking Mobile App**

*Flutter, Dart, Firebase Authentication, Cloud Firestore, Provider, ChangeNotifier, Material 3, fl_chart*

- Developed a Flutter-based Android fitness and nutrition tracking application with Firebase Authentication supporting email/password sign-up, login, password reset, logout, and guest mode.
- Designed an AI-coach-style dashboard displaying fitness goals, calorie tracking, macronutrient breakdowns, hydration progress, recovery and streak cards, and daily workouts using Material 3 components.
- Integrated Cloud Firestore with a structured users/{uid} data model to persist user profiles, selected goals, meal logs, water intake records, and workout completion data.
- Implemented workout, nutrition, hydration, progress analytics, and profile modules using Provider and ChangeNotifier for state management with reusable UI components.

**Leaf Lab - Plant Disease Detection Mobile App**

*TensorFlow, TensorFlow Lite, Flutter, CNN, Data Augmentation*

- Trained a CNN classifier on 87,000 augmented images spanning 38 plant disease classes using TensorFlow and offline data augmentation techniques.
- Achieved 97.3% test accuracy with balanced Precision and Recall of 97.3% across all classes.
- Optimized the model with TensorFlow Lite quantization for on-device mobile inference and deployed on Flutter, enabling real-time disease detection without internet connectivity.

**Medical Image Segmentation - Breast Ultrasound Analysis Web App**

*U-Net, CNN, MongoDB, Python, Full-Stack Web Development*

- Built a full-stack web application for automated segmentation and classification of breast ultrasound tumors into normal, benign, and malignant categories.
- Trained U-Net and CNN models on 780 ultrasound images from 600 patients, achieving an F1-score of 0.85.
- Independently implemented the end-to-end pipeline including model training, web deployment, and MongoDB integration for persistent data storage.

**Restaurant Chatbot - Multi-Turn RAG Assistant**

*LangChain, RAG, FAISS, WhatsApp API, Embeddings*

- Developed a RAG-powered WhatsApp chatbot for restaurant operations, integrating menu and ingredient databases for context-aware recommendations.
- Implemented semantic retrieval using FAISS vector database and text embeddings to support multi-turn conversations covering menu queries, ingredient substitutions, and dietary preferences.

**Real-Time Sentiment Analysis Dashboard**

*Transformers, Python, NLP, Data Visualization*

- Built a real-time sentiment classification pipeline using transformer-based NLP models applied to live scraped tweet data.
- Developed an interactive dashboard for monitoring and visualizing sentiment trends over time.

**YOLOv5 Object Detection System**

*YOLOv5, OpenCV, Python, Custom Dataset*

- Trained a YOLOv5 model on a custom-annotated dataset for multi-class object recognition.
- Deployed real-time object detection on live webcam streams using OpenCV.

## ACHIEVEMENTS

- Led a 5-member team to deliver an AI and Mixed Reality graduation project (ARAGbMR) that won project funding of 52,250 EGP and was successfully showcased at AASTMT University.
- Presented an applied RAG and AR educational system during formal university evaluation, receiving an engagement score of 4.6 out of 5.

## CERTIFICATIONS

- Machine Learning (60 hrs) - Information Technology Institute (ITI), Egypt | Aug-Sep 2024 - Covered Supervised Machine Learning, Unsupervised Machine Learning, and Python Libraries for Machine Learning.

## LANGUAGES

**Arabic:** Native

**English:** Professional